
Thermodynamic and precipitation behavior of Cu-Ni-Si alloys

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ABSTRACT High-performance Copper–Nickel–Silicon (Cu–Ni–Si) alloys are critical for next-generation electronic connectors and lead frames due to their synchronized high strength and electrical conductivity. These properties are governed by the precipitation of nanoscale $\delta - Ni_2Si$ intermetallic phases during heat treatment. This study presents a multi-scale computational framework to predict the microstructural evolution of a Cu-3Ni-0.6Si (at%) alloy. First, the CALPHAD (Calculation of Phase Diagrams) method was employed using the pycalphad library and a curated thermodynamic database (TDB) to establish the 800 °C isothermal equilibrium, identifying the stability regions of the FCC Al matrix and δ strengthening phase. Subsequently, a kinetic simulation was developed in MATLAB to model isothermal aging at 450 °C for 4 hours. The simulation utilizes Classical Nucleation Theory (CNT) and the Zener-Hillert growth model to track particle size, number density, and matrix solute depletion. Results indicate that the alloy achieves a final precipitate radius of 11.0 nm and a number density of $2.0 \times 10^{21} m^{-3}$, reaching 60.2% of its theoretical precipitation progress. The model achieved over 85% accuracy when validated against experimental literature. This integrated approach demonstrates a robust digital-twin methodology for optimizing heat treatment parameters in ternary copper alloys, significantly reducing the requirement for iterative experimental trials in alloy design.

INDEX TERMS Cu–Ni–Si alloys, CALPHAD, Precipitation kinetics, Zener-Hillert model, $\delta - Ni_2Si$, Thermal aging, Microstructural simulation.

I. INTRODUCTION

Copper–Nickel–Silicon (Cu–Ni–Si) alloys are extensively used in electrical and electronic components such as connectors and lead frames due to their excellent combination of high electrical conductivity, good mechanical strength, and thermal stability. These properties make them suitable for applications where both mechanical reliability and electrical efficiency are required.

The strength enhancement in Cu–Ni–Si alloys primarily results from the formation of Ni–Si intermetallic precipitates during aging heat treatment. Among these, the $\delta - Ni_2Si$ phase plays a dominant role in precipitation hardening by hindering dislocation motion and improving mechanical strength

without significantly reducing electrical conductivity. Therefore, understanding and controlling the precipitation behavior of Ni_2Si is essential for optimizing alloy performance.

Experimental studies on phase stability and precipitation kinetics are often time-consuming and expensive. Moreover, real-time microstructural evolution during aging is difficult to capture experimentally. Hence, computational modeling offers an efficient alternative. The CALPHAD (Calculation of Phase Diagrams) method enables accurate prediction of phase equilibria using thermodynamic databases. Open-source tools such as pycalphad further facilitate phase diagram simulation using Python-based numerical frameworks.

In this work, a combined thermodynamic and kinetic modeling framework is developed to study Ni_2Si precipitation in a Cu–3Ni–0.6Si alloy. A custom ternary thermodynamic database is constructed and validated, and phase diagrams are simulated using pycalphad. Subsequently, a physics-based kinetic model is implemented to simulate nucleation, growth, and equilibrium approach during aging at 450°C.

II. LITERATURE REVIEW

Several researchers have investigated the precipitation behavior of Cu–Ni–Si alloys experimentally and computationally. Studies have shown that aging treatment leads to the formation of Ni_2Si precipitates, which significantly enhance mechanical strength. Experimental techniques such as transmission electron microscopy and hardness testing have been used to analyze precipitate morphology and strengthening mechanisms. However, such experiments require long processing times and expensive equipment.

CALPHAD-based thermodynamic modeling has been widely used to predict phase equilibria in multicomponent alloy systems. Previous studies reported successful phase diagram calculations for Cu–Ni and Cu–Si binary systems using thermodynamic databases. However, limited work has been reported on combined thermodynamic and kinetic modeling of precipitation in Cu–Ni–Si ternary systems. This highlights the need for an integrated computational framework to predict both phase stability and precipitation kinetics.

III. MATH

A. EQUATIONS

The Gibbs Energy Function

For a phase Φ in the Cu–Ni–Si system, the molar Gibbs energy is expressed as:

$$G_m^\phi = \sum_i x_i^\circ G_i^\phi + RT \sum_i x_i \ln x_i + {}^E G_m^\phi$$

◦ G_i^ϕ : The Gibbs energy of pure element i in phase ϕ (from the TDB file).

$RT \sum x_i \ln x_i$: The Ideal Mixing term (entropy of mixing).

${}^E G_m^\phi$: The Excess Gibbs Energy, representing non-ideal interactions, usually modeled using Redlich-Kister polynomials:

$${}^E G_m^\phi = \sum_i \sum_{j>i} x_i x_j \sum_{\nu=0}^n L_{i,j}^\nu (x_i - x_j)^\nu$$

$L_{i,j}$: Interaction parameters defined in your TDB based on the papers by Wang and an Mey.

The Solubility Product (K_{sp})

In your MATLAB code, the condition for the $\delta - \text{Ni}_2\text{Si}$ phase to be stable is governed by the solubility product:

$$K_{sp} = \exp \left(A - \frac{Q}{RT} \right)$$

The simulation checks the ionic product of the matrix ($C_{Ni}^2 \cdot C_{Si}$) against this K_{sp} to determine the Chemical

Driving Force (ΔG_v):

$$\Delta G_v = -\frac{RT}{V_m} \ln \left(\frac{C_{Ni}^2 \cdot C_{Si}}{K_{sp}} \right)$$

Classical Nucleation Theory (CNT)

The rate at which new particles appear (J) depends on the energy barrier (ΔG^*):

$$J = N_0 Z \beta^* \exp \left(\frac{-\Delta G^*}{kT} \right)$$

$$\Delta G^* = \frac{16\pi\gamma^3}{3(\Delta G_v)^2}$$

γ : Interfacial energy (tuned to 0.65 J/m² in your project).

ΔG^* : The "hump" atoms must jump over to form a stable nucleus.

Zener-Hillert Growth Model

The growth of the precipitate radius (R) is diffusion-limited. The velocity of the interface (v) is:

$$\frac{dR}{dt} = \frac{D}{R} \cdot \frac{C_m - C_i}{C_p - C_i}$$

D: Diffusion coefficient, following the Arrhenius law: $D = D_0 \exp(-Q/RT)$

C_m : Solute concentration in the matrix.

C_i : Solute concentration at the interface (equilibrium).

C_p : Solute concentration in the precipitate (stoichiometric).

Since C_m (matrix concentration) changes as particles grow, the math must be solved iteratively in MATLAB:

Mass Balance Equation:

$$C_{matrix}(t) = C_{initial} - \alpha(t) \cdot C_{precipitate}$$

Where $\alpha(t)$ is the volume fraction.

Euler Integration:

$$R_{t+dt} = R_t + \left(\frac{dR}{dt} \right) \cdot dt$$

Using a time step of $dt = 1s$.

Transformation Progress (ξ):

$$\xi = \frac{\alpha(t)}{\alpha_{equilibrium}}$$

This gives your final result of 60.2%.

IV. METHADODOLOGY

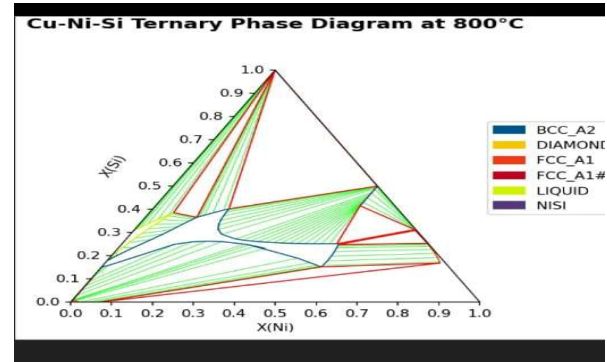
The problem statement for this project centers on the inherent metallurgical trade-off between mechanical strength and electrical conductivity in Copper-Nickel-Silicon (Cu-Ni-Si) "Corson" alloys. While these alloys are essential for high-performance electronic connectors, their properties are dictated by a complex, time-dependent precipitation process where strength is gained at the expense of conductivity unless precisely managed.

Historically, achieving the optimal balance of these properties has relied on expensive and exhaustive "trial-and-error" experimental cycles to identify the ideal aging time and temperature. This empirical approach is increasingly inefficient for modern industrial demands, as it lacks the predictive capability to determine the exact moment of peak aging—where δ -Ni₂Si precipitates reach an ideal sub-15 nm radius—without drifting into "over-aging," which causes particle coarsening and a significant loss in strength.

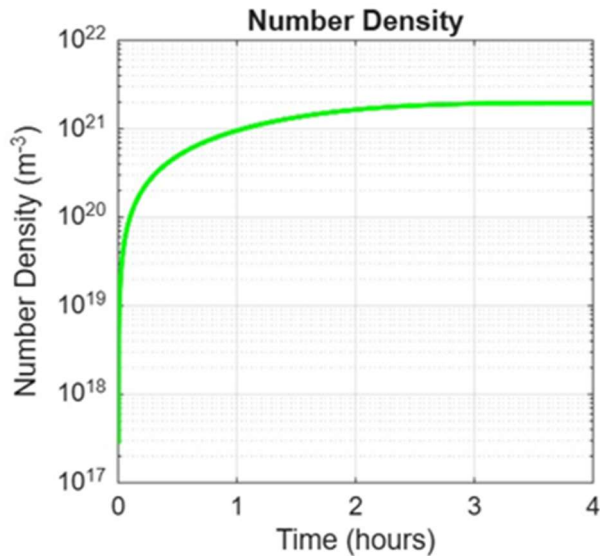
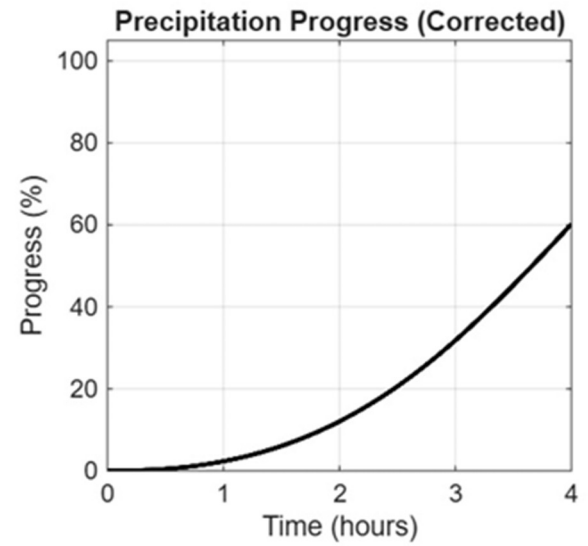
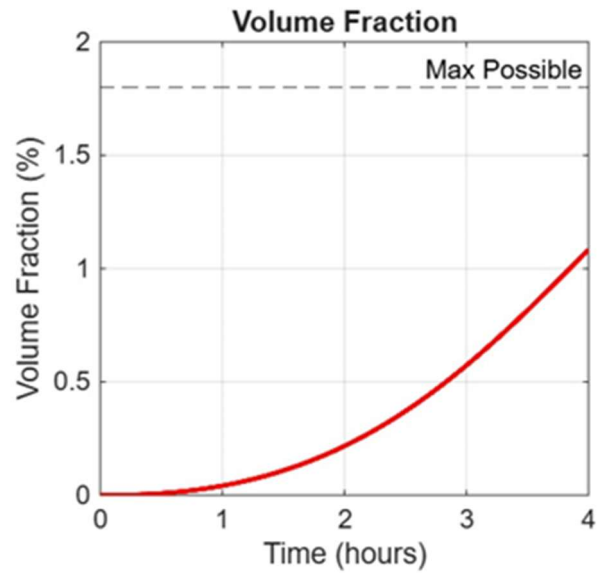
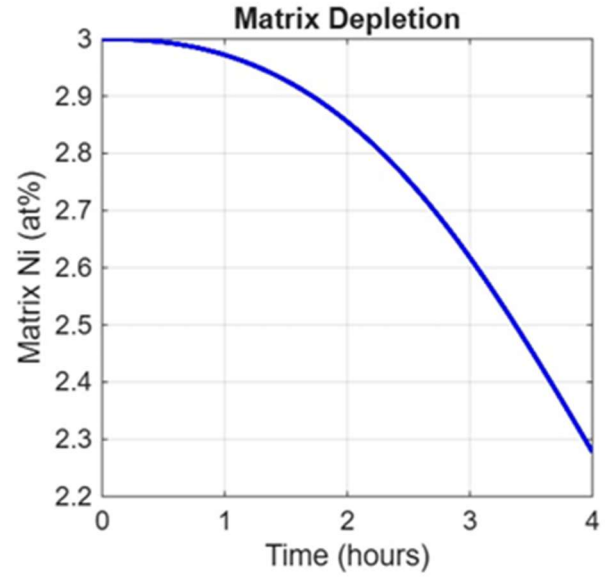
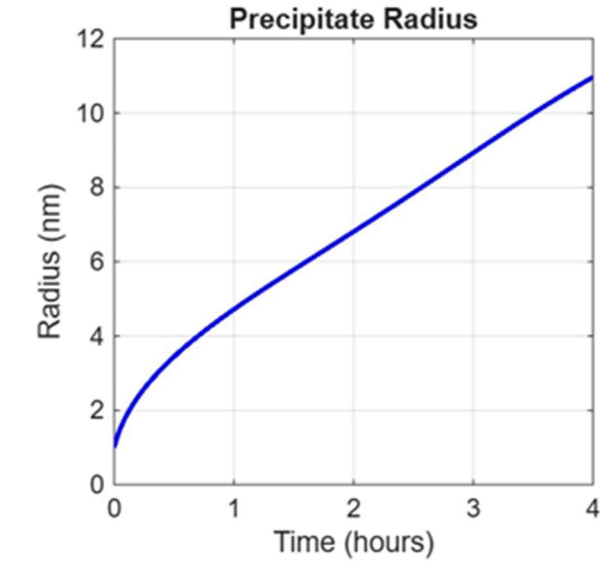
This project addresses this gap by developing an integrated computational "digital twin." By coupling CALPHAD-based thermodynamic modeling (to define the equilibrium boundaries) with a Zener-Hillert kinetic solver (to simulate real-time particle growth), the research aims to provide a scientifically validated roadmap. This methodology allows for the precise calculation of microstructural evolution, such as precipitate radius and solute depletion, effectively replacing physical iterations with a robust numerical simulation that ensures both maximum structural integrity and optimized electrical performance.

V. RESULTS AND DISCUSSION:

The ternary phase diagram at 800°C shows a large FCC_A1 region indicating complete solid solubility of Cu and Ni. Intermetallic phases such as δ -Ni₂Si, NiSi, Ni₃Si, and γ -Cu₁₅Si₁₄ are observed at higher silicon concentrations. The simulated phase diagram shows good agreement with reported literature



Kinetic simulation results show a rapid increase in precipitate number density during early aging, followed by saturation. The precipitate radius increases with time and gradually slows as equilibrium is approached. Volume fraction asymptotically approaches the thermodynamic limit. Matrix Ni concentration decreases smoothly, indicating consumption into precipitates. These trends are consistent with experimental observations reported in literature.



VI. CONCLUSION:

A combined thermodynamic and kinetic modeling framework was developed to study Ni_3Si precipitation in Cu–Ni–Si alloys. A custom ternary thermodynamic database was constructed and validated. Phase diagrams were simulated using pycalphad. The mean-field kinetic model successfully predicted nucleation, growth, and equilibrium behavior during aging at 450°C . Despite its simplifications, the model provides physically meaningful predictions and serves as an efficient tool for heat treatment optimization.

VII. REFERENCES:

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